Modelling of extreme excursions of stochastic processes in space and time

Foundations for a new tail process

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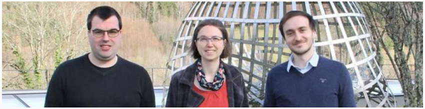
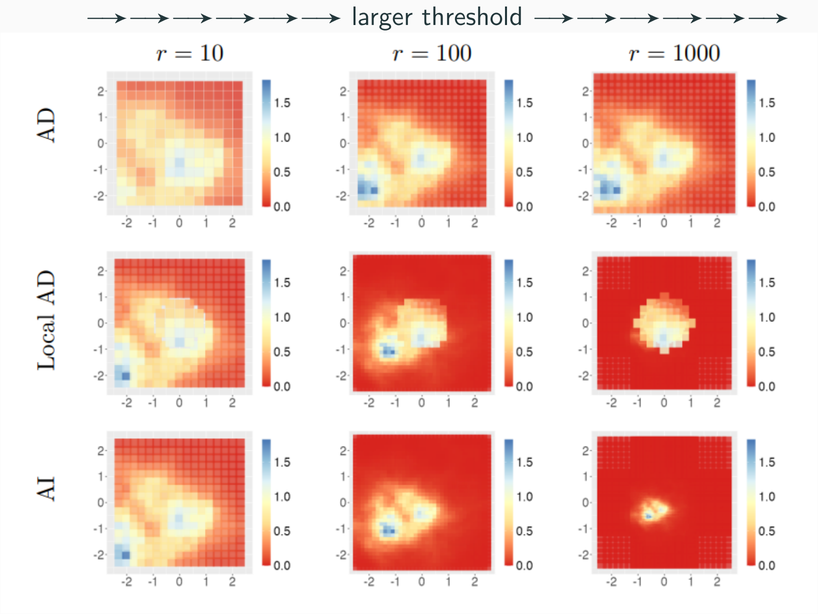
EXtremes and Rare Events in Climate and related applications
University of Reading - June 2025

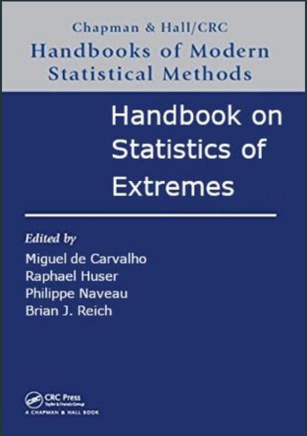
Joint work with Marco Oesting and Raphaël de Fondeville



How do extreme events look like in your data sets?



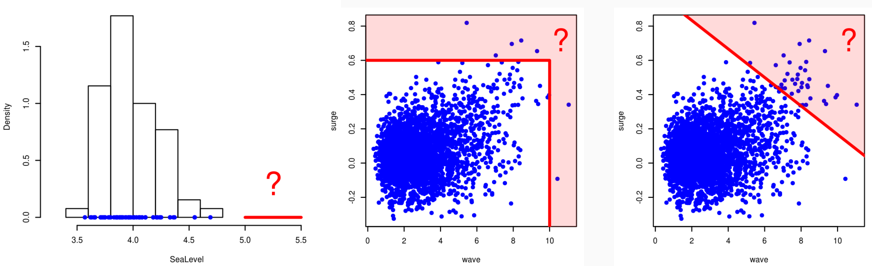
Background on (Spatial/Temporal) Extreme Value Analysis



Motivation

Extreme value theory

= theoretical basis for extrapolation into the regime of rare events

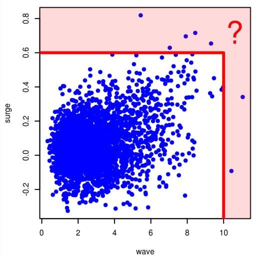


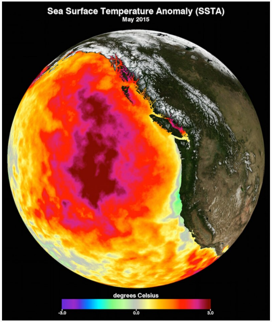
data portpirie and wavesurge from R package ismev

Typical applications

- hydrology (e.g. dikes), meteorology (wind gusts, heavy rainfall)
- finance/insurance (reserve funds to cover big losses)

Extreme events of 2 variables, 1000 variables, temporal/spatial footprint, ...



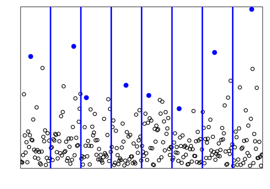


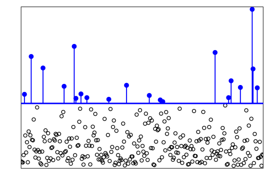
By NASA Physical Oceanography Distributed Active Archive Center - Laura Naranjo "The Blob", March 15th 2019 https://commons.wikimedia.org/w/index.php?

Inappropriate: models for fluctuations around the mean
EVT approach: models from limit theorems/regularity at the boundary

Univariate "vanilla" approaches

Consider repeated measurements of 1 variable (e.g. temperature at 1 location)





Block maxima $(M_j)_{j=1,2,\ldots,r}$
$\approx$ Sample from max-stable distribution (GEV)

$$G_\gamma\left(\frac{x-\mu}{\sigma}\right)$$

with $G_\gamma(x) = \exp\{-(1+\gamma x)_+^{-1/\gamma}\}$

Threshold-Exceedances $X_j - t \mid X_j > t$
$\approx$ Sample from threshold-stable distribution (GPD)

$$H_\gamma\left(\frac{x}{\sigma}\right)$$

with $H_\gamma(x) = \{1-(1+\gamma x)_+^{-1/\gamma}\}_+$

A. Bücher and C. Zhou. A horse racing between the block maxima method and the peak-over-threshold approach. *Statistical Science* 36(3):360-378, 2021.

Mathematical foundation

Let $X_1, X_2, X_3, \ldots$ be independent copies of X.
Under appropriate normalizations, the following are **equivalent**:

$$\frac{\max(X_1, X_2, \ldots, X_n) - b_n}{a_n} \longrightarrow G = \exp(-\overline{\Lambda}) \qquad \textbf{Maxima cvg.}$$

$$\left\{ \frac{X_1 - b_n}{a_n}, \frac{X_2 - b_n}{a_n}, \ldots, \frac{X_3 - b_n}{a_n} \right\} \longrightarrow \text{Poisson point process}(\Lambda) \qquad \textbf{Point process cvg.}$$

$$n \,\mathrm{Prob}\left(\frac{X - b_n}{a_n} \in \cdot \right) \longrightarrow \Lambda \qquad \textbf{Regular variation}$$

$$\frac{1}{\mathrm{Prob}(X > u)} \mathrm{Prob}\left(\frac{X - u}{\sigma(u)} \in \cdot \right) \longrightarrow \Lambda$$

$$\left. \frac{X - u}{\sigma(u)} \right| X > u \longrightarrow H = (1 - \overline{\Lambda})_+ \qquad \textbf{Exceedance cvg.}$$

The limiting objects have **stability properties (self-similarities)**.

Spatial/Temporal extensions (so far)

- Replace random variable X by random function $X = \{X(s)\}_{s \in S}$.

  (random field, stochastic process, ...)

The following are **equivalent** (on standard Pareto marginal scale):

$$\max(X_1, X_2, \dots, X_n)/n \longrightarrow \text{max-stable process } Z \text{ (with exponent measure } \Lambda)$$

$$\{X_1/n, X_2/n, \dots, X_n/n\} \longrightarrow \text{Poisson point process (with intensity measure } \Lambda)$$

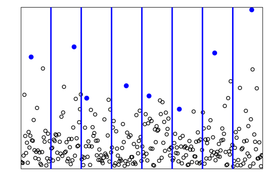
$$n \,\mathrm{Prob}\big(X/n \in \cdot\big) \longrightarrow \Lambda = (\nu_1 \otimes \mathcal{L}(V)) \circ m^{-1}$$

And **imply**:

$$\frac{X}{n} \,\Big|\, X(s_0) > n \longrightarrow \text{Pareto process } \{PV^{(s_0)}(s)\}_{s \in S}$$

C. Dombry and M. Ribatet. Functional regular variations, Pareto processes and peaks over threshold. *Stat. Interface* 8(1):9-17, 2015.

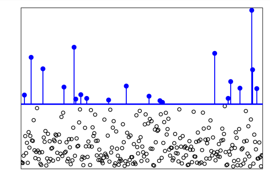
Implications for statistical practice



Block maxima

$$\{M_j(s_i)\}_{i,j}$$

$\approx$ Sample from
max-stable process
at sites s_i



Threshold-Exceedances

$$\{X_j(s_i)\} \mid X_j(s_0) > t$$

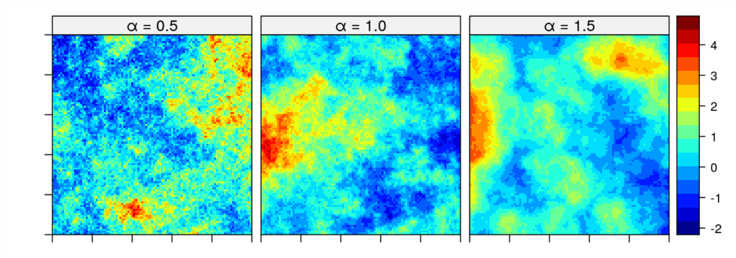
$\approx$ Sample from
(generalized) Pareto process
at sites s_i

Both sides (max-stable and Pareto processes) can be **decomposed** into

- marginal information (GEV/GPD) and
- dependence information ("spectral" processes $V^{(s_0)}$).

Popular model for dependence

Brown-Resnick process (max-stable process)



Corresponding Pareto process:

$$Y^{(s_0)}(s) = \underbrace{\underbrace{P}_{\text{size of event}} \quad \underbrace{V^{(s_0)}(s)}_{\substack{\text{spatial profile}\\ \text{around } s_0}}}_{\text{independent}}, \quad \text{where} \quad V^{(s_0)}(s) = \exp\Big\{ \underbrace{W^{(s_0)}(s)}_{\substack{\text{Gaussian process}\\ \text{with variogram } \gamma \\ \text{e.g. } \gamma(s,s_0) \\ = \|s-s_0\|^{\alpha} \\ \text{(fractional Brownian field)}}} - \gamma(s, s_0) \Big\}$$

$=$ resulting model for conditioning on being large at s_0 (on Pareto scale)

Recent applications of Pareto processes

Spatio-temporal analysis of extreme winter temperatures in Ireland:

D. Healy, J. Tawn, P. Thorne and A. Parnell. Inference for extreme spatial temperature events in a changing climate with application to Ireland. *J. R. Stat. Soc. Ser. C.* qlae047, 2024.

Windspeed and rainfall:

R. de Fondeville and A.C. Davison. Functional peaks-over-threshold analysis *J. R. Stat. Soc. Ser. B.* 84(4): 1392-1422, 2022.

Windspeed: Storm Daria 1990 (de Fondeville and Davison 2022)

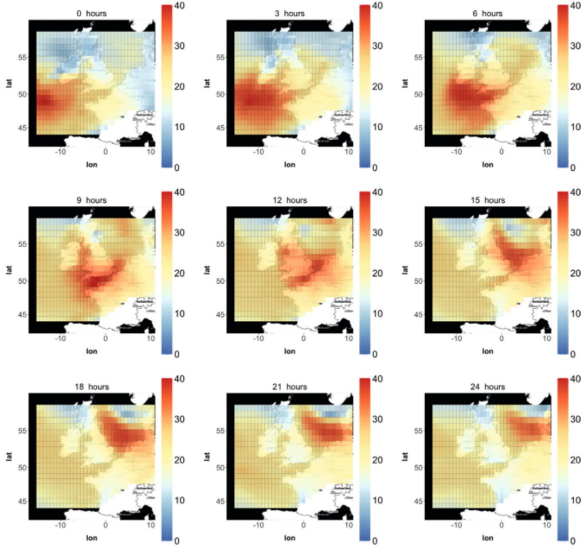


FIGURE 4 Maximum speed (ms^{-1}) over the past 3 h of the wind gusts sustained for at least 3s from ERA-Interim reanalysis during the peak of windstorm Daria, which swept over Europe during January 1990 [Colour figure can be viewed at wileyonlinelibrary.com]

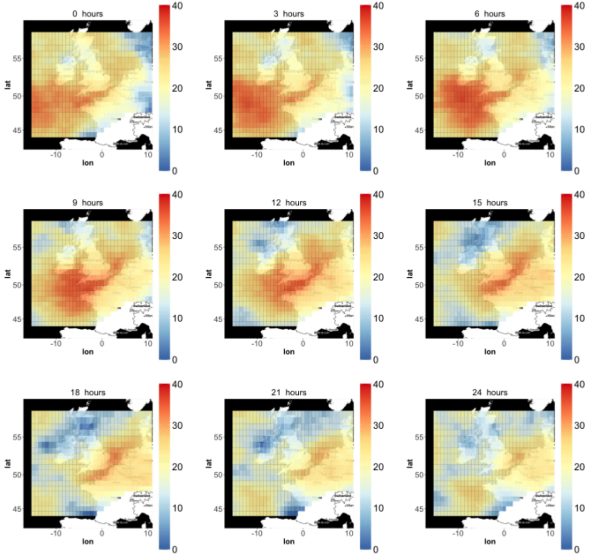


Fig. 8. Simulated maximum speed (ms^{-1}) over the past $3h$ hours of wind gusts sustained for at least 3s. The storm has an intensity $r(x) = 29.1\ \ \mathrm{ms}^{-1}$.

However ... stability (self-similarity) very rigid

no link

between **size** of extreme event and its (law of) **spatial profile**

$\downarrow$

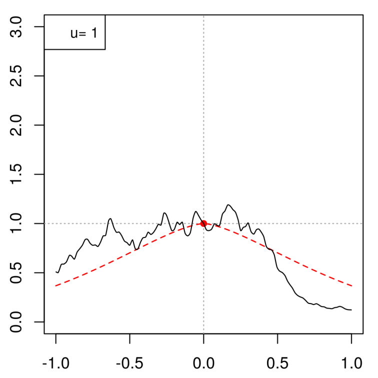
Pareto processes can only model

"asymptotic dependence (AD)"

Samples from

$$Y^{(0)} \mid Y^{(0)}(0) = u$$

for $u = \dots$



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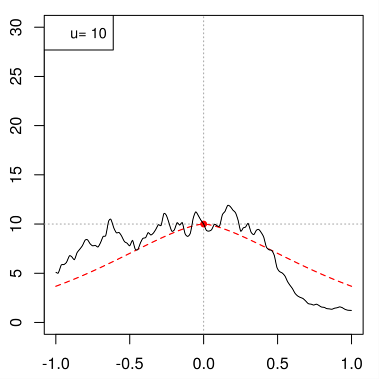

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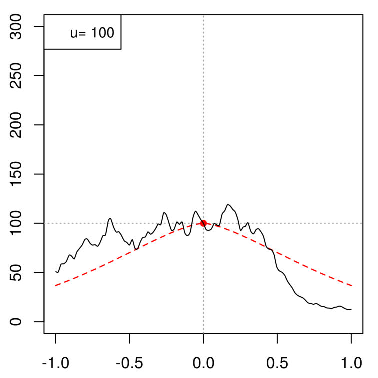

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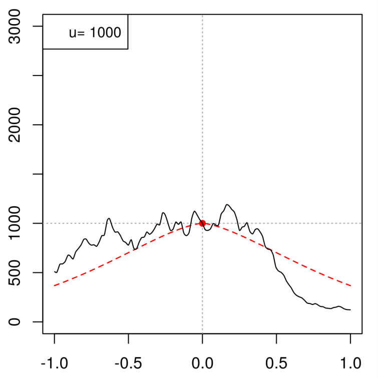

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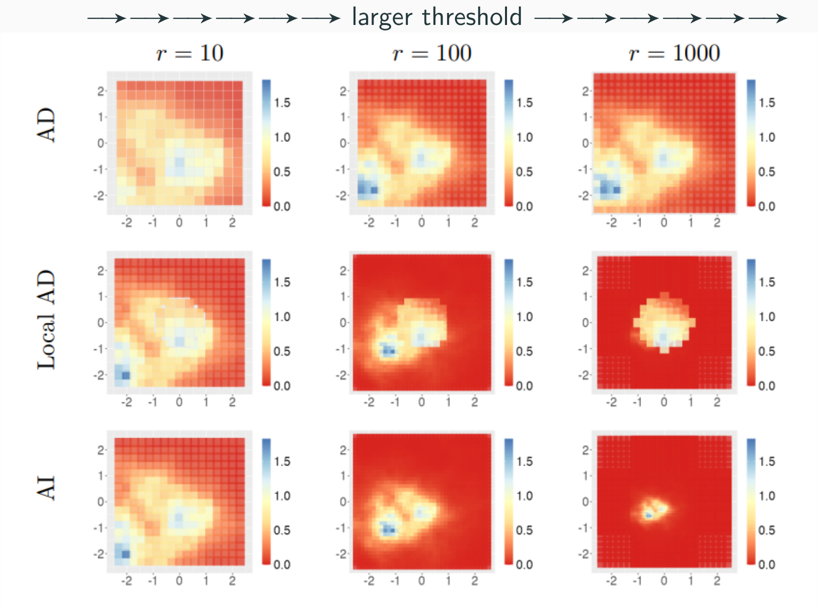

Basic practical demands

A model for space-time footprints of extreme events should:

- reflect **single rather than composite extreme events**
- be flexible to allow for **modelling across different dependency regimes** known as asymptotic dependence (AD) and asymptotic independ'ce (AI),
- be **consistent across different spatio-temporal conditioning**.

So far only partially met:

property / *model approach*	event based	dependence regimes: weak (AI)	dependence regimes: strong (AD)	consistency guarantee
Max-stable process	✗	✗	✓	✓
Inverted max-stable process	✗	✓	✗	✓
Max-infinitely div. process	✗	✓	✓	✓
Pareto process	✓	✗	✓	✓
Conditional extremes	✓	✓	✓	✗
AI/AD product mixture	✗	✓	✓	✓

No universal principle!

Idea: Include spatial scaling in stability

Example: A different maxima convergence:

Let $X_1, X_2, X_3, \ldots$ be independent copies of a Gaussian process X.
Under appropriate normalizations:

$$\underbrace{\max_{k=1,\ldots,n} \left(\frac{X_k(\,\cdot\,/\delta_n) - b_n}{a_n} \right)}_{\text{asymptotic independent (AI)}} \longrightarrow \underbrace{\textbf{Brown-Resnick process}}_{\text{asymptotic dependent (AD)}}$$

Z. Kabluchko, M. Schlather and L. de Haan. Stationary max-stable fields associated to negative definite functions. *Ann. Probab.* 37(5):2042-2065, 2009.

Idea: investigate threshold exceedance (and other) counterparts.

New mathematical foundation including spatial scaling

Let $X_1, X_2, X_3, \ldots$ be independent copies of a (not necessarily Gaussian!) process X (on Pareto scale).

The following are **equivalent**:

$$\max\left\{\frac{X_1(\,\cdot\,/\delta_n)}{n}, \frac{X_2(\,\cdot\,/\delta_n)}{n}, \ldots, \frac{X_n(\,\cdot\,/\delta_n)}{n}\right\} \longrightarrow \text{Brown-Resnick process}$$

$$\left\{\frac{X_1(\,\cdot\,/\delta_n)}{n}, \frac{X_2(\,\cdot\,/\delta_n)}{n}, \ldots, \frac{X_n(\,\cdot\,/\delta_n)}{n}\right\} \longrightarrow \begin{array}{l}\text{corresponding}\\ \text{Poisson point process}\end{array}$$

$$n\,\mathrm{Prob}\left(\frac{X(\,\cdot\,/\delta_n)}{n} \in \cdot\right) \longrightarrow \begin{array}{l}\text{corresponding}\\ \text{limit measure}\end{array}$$

And **imply**:

$$\frac{X(\,\cdot\,/\delta_n)}{n}\,\Bigg|\, X(0) > n \longrightarrow \begin{array}{l}\text{corresponding}\\ \text{Pareto process}\end{array}$$

Examples: X Brown-Resnick process (AD) $\rightarrow \delta_n \equiv 1$

X Gaussian process (AI) (on Pareto scale) $\rightarrow \delta_n \sim c\log(n)^{1/\alpha}$

with correlation $\rho(h) \sim 1 - \|h\|^\alpha$, $\alpha \in (0, 2)$

New tail process (instead of Pareto process)

Old tail process:

$$Y^{(0)}(s) = PV^{(0)}(s)$$

New tail process:

$$Y^{(0)}(s) = PV^{(0)}(\delta(P)s)$$

---

now link

between **size** of extreme event
and its (law of) **spatial profile**

$\downarrow$

New model can straddle
"AI/AD behaviour"

Samples from

$$Y^{(0)} \mid Y^{(0)}(0) = u$$

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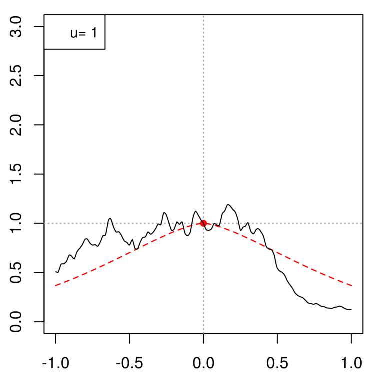

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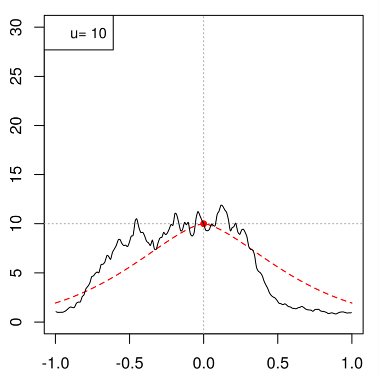

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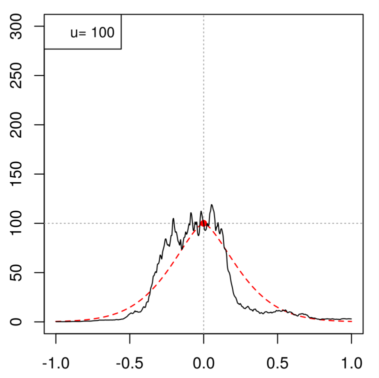

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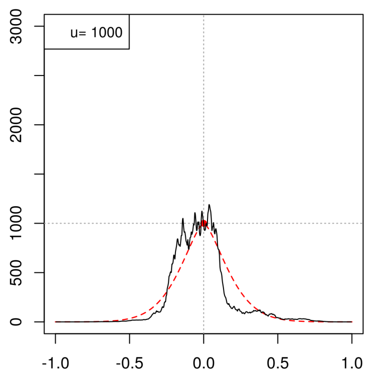


Local AD

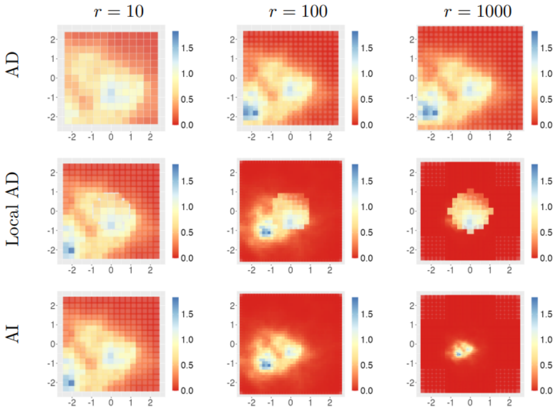


Figure 3: Simulations of the angular process $V^{X(0,0)}$ using three examples of grid deformations at intensity levels $R = 1, 10, 100$. First row: $\delta_1(R, s) = [2 - \exp\{-(R-1)/20\}]s$. Last row : $\delta_2(R, s) = (1 + \sqrt{R-1}/4)s$ Center row: the deformation function is defined through δ_1 if $|s| \leq 1$ and δ_2 if $|s| > 1$.

Example

New tail process

$$Y^{(0)}(s) = PV^{(0)}(\delta(P)s)$$

with

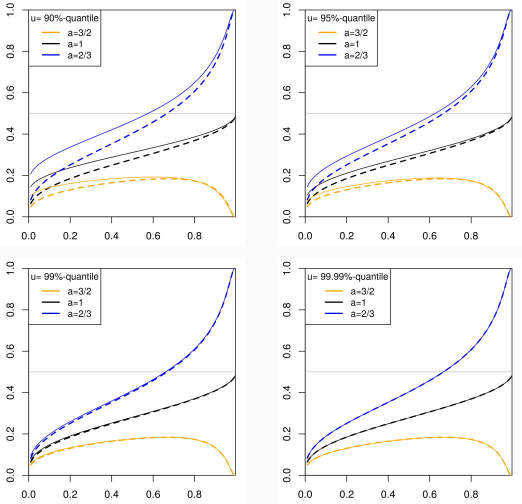
$\log V^{(0)}$ α-fractional Brownian field (with trend)

and

$$\delta(u) \sim \log(u)^{\beta/\alpha}$$

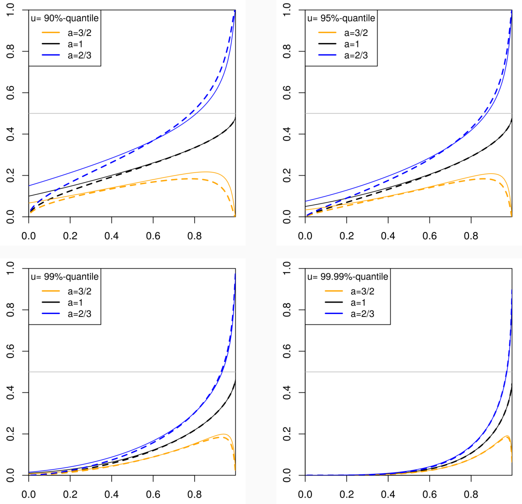
$\beta = 0$	$\beta = 1$
$\downarrow$	$\downarrow$
$Y^{(0)}$ approximates tails of Brown-Resnick processes (and similar)	$Y^{(0)}$ approximates tails of Gaussian processes (and similar)
(asymptotic dependent - AD)	(asymptotic independent - AI)

Approximation of conditional exceedance probabilities ($\beta = 0$)



Conditional exceedance probabilities $\mathbb{P}(X(s) > au \,|\, X(s_0) = u)$ of the Brown–Resnick process from (solid) versus approximations $\mathbb{P}(Y^{(s_0)}(s) > au \,|\, Y^{(s_0)}(s_0) = u)$ from the tail process (dashed) as functions of $\exp(-\gamma) \in (0, 1)$ for different values of u and a.

Approximation of conditional exceedance probabilities $(\beta = 1)$



Conditional exceedance probabilities $\mathbb{P}(X(s) > au \,|\, X(s_0) = u)$ of the Gaussian copula process (solid) versus approximations $\mathbb{P}(Y^{(s_0)}(s) > au \,|\, Y^{(s_0)}(s_0) = u)$ from the tail process (dashed) as functions of $\rho = \exp(-\gamma) \in (0, 1)$ for different values of u and a.

Conclusions

Thank you!

- extreme value analysis = theoretical basis for (statistical) extrapolation into the regime of rare events
- classical spatial extremes models have **too rigid stability properties**
- **our approach:** new form of stability (self-similarity) including spatial scaling that can model localization of extremes and straddles AI/AD
- for now: **mathematical foundation:**
  Preprint: https://publications.mfo.de/handle/mfo/4206

Next?

- comparison with other standard models (M. Kipping (Stuttg.))
- explore connection with "On the spatial extent of extreme threshold exceedances" (Cotsakis, DiBernadino, Opitz)
- statistical inference / properties / guidance / case studies !!!